



LESSON PLAN	COURSE: MDM4U
Date:	Unit of Study: Unit 4
Materials/Resources: Mathmetics of data management study guide	Lesson Focus/Theme: Chapter 8: The Normal Distribution
Prior Knowledge: Probability	
Overall Expectations: By the end of this lesson, students will: describe the characteristics of a good sample, some sampling techniques, and principles of primary data collection, and collect and organize data to solve a problem.	

Specific Expectations:

2.1 determine and describe principles of primary data collection (e.g., the need for randomization, replication, and control in experimental studies; the need for randomization in sample surveys) and criteria that should be considered in order to collect reliable primary data (e.g., the appropriateness of survey questions; potential sources of bias; sample size)

2.2 explain the distinction between the terms population and sample, describe the characteristics of a good sample, explain why sampling is necessary (e.g., time, cost, or physical constraints), and describe and compare some sampling techniques (e.g., simple random, systematic, stratified, convenience, voluntary)

2.3 describe how the use of random samples with a bias (e.g., response bias, measurement bias, non-response bias, sampling bias) or the use of non-random samples can affect the results of a study

2.4 describe characteristics of an effective survey (e.g., by giving consideration to ethics, privacy, the need for honest responses, and possible sources of bias, including cultural bias), and design questionnaires (e.g., for determining if there is a relationship between a person's age and their hours per week of Internet use, between marks and hours of study, or between income and years of education) or experiments (e.g., growth of plants under different conditions) for gathering data

2.5 collect data from primary sources, through experimentation, or from secondary sources (e.g., by using the Internet to access reliable data from a well-organized database such as E-STAT; by using print sources such as newspapers and magazines), and organize data with one or more attributes (e.g., organize data about a music collection classified by artist, date of recording, and type of music using dynamic statistical software or a spreadsheet) to answer a question or solve a problem

Learning Goals:

By the end of this lesson, students will know the normal approximation to the binomial distribution and confidence interval

Success Criteria:

By the end of this lesson, I will be able to know the normal approximation to the binomial distribution

By the end of this lesson, I will be able to know the confidence intervals

Assessment Strategies: Tests, quiz and project

Time:
10 min

Introduction/Minds-On:

Answer questions students don't understand from last class if any

Time: 80 min	Lesson: Chapter 8: The Normal Distribution 8.4 Normal approximation to the binomial distribution 8.5 Confidence intervals
Time: 20min 10 min	Extension / Wrap-up: Review the key concepts and formula for whole course Do presentation on the project and then hand in the report
Notes / Reminders / Homework: Do test #4 before or after the class in classroom for one hour	
Lesson Reflection:	